

# Kotlin Collections Practice - Beginner + Intermediate (60 Questions)

## Beginner (20 Questions)

### **1. Print all elements of a list**

Input:

```
val nums = listOf(1, 2, 3, 4, 5)
```

Output:

```
1
2
3
4
5
```

### **2. Print first and last element**

Input:

```
val nums = listOf(10, 20, 30, 40, 50)
```

Output:

First: 10

Last: 50

### **3. Reverse a list**

Input:

```
val nums = listOf(1, 2, 3, 4, 5)
```

Output:

```
[5, 4, 3, 2, 1]
```

### **4. Mutable list add/remove**

Input:

```
val fruits = mutableListOf("Apple", "Banana", "Orange")
```

Action: Add "Grapes", "Mango", remove "Banana"

Output:

```
[Apple, Orange, Grapes, Mango]
```

### **5. Check if list contains number**

Input:

```
val nums = listOf(5, 10, 15, 20)
```

Check if 15 exists

Output:

```
true
```

### **6. Set with duplicates**

Input:  
val nums = listOf(1, 2, 2, 3, 3, 4, 5)  
Convert to Set  
Output:  
[1, 2, 3, 4, 5]

### **7. Find size of a set**

Input:  
val nums = setOf(1, 2, 3, 4, 5)  
Output:  
5

### **8. Convert list → set → list**

Input:  
val nums = listOf(1, 1, 2, 2, 3, 3, 4)  
Output:  
[1, 2, 3, 4]

### **9. Map of countries → capitals**

Input:  
val capitals = mapOf("USA" to "Washington", "India" to "New Delhi", "Japan" to "Tokyo")  
Print India's capital  
Output:  
New Delhi

### **10. Add entry to mutable map**

Input:  
val capitals = mutableMapOf("France" to "Paris", "Germany" to "Berlin")  
Add Spain → Madrid  
Output:  
{France=Paris, Germany=Berlin, Spain=Madrid}

### **11. Remove key from map**

Input:  
val capitals = mutableMapOf("UK" to "London", "Italy" to "Rome")  
Remove UK  
Output:  
{Italy=Rome}

### **12. Check if map contains key**

Input:  
val capitals = mapOf("China" to "Beijing", "Russia" to "Moscow")  
Check if China exists  
Output:  
true

### **13. Check if map contains value**

Input:

```
val capitals = mapOf("Brazil" to "Brasília", "Argentina" to "Buenos Aires")
```

Check if Brasília exists

Output:

true

### **14. Get all keys of a map**

Input:

```
val capitals = mapOf("USA" to "Washington", "India" to "New Delhi", "Japan" to "Tokyo")
```

Output:

[USA, India, Japan]

### **15. Get all values of a map**

Input:

```
val capitals = mapOf("USA" to "Washington", "India" to "New Delhi", "Japan" to "Tokyo")
```

Output:

[Washington, New Delhi, Tokyo]

### **16. Convert strings to uppercase**

Input:

```
val words = listOf("apple", "banana", "cherry")
```

Output:

[APPLE, BANANA, CHERRY]

### **17. Sort list ascending**

Input:

```
val nums = listOf(50, 10, 30, 20, 40)
```

Output:

[10, 20, 30, 40, 50]

### **18. Sort list descending**

Input:

```
val nums = listOf(5, 2, 8, 1, 3)
```

Output:

[8, 5, 3, 2, 1]

### **19. Find length of a list**

Input:

```
val fruits = listOf("Apple", "Banana", "Cherry")
```

Output:

3

### **20. Join strings with commas**

Input:  
val words = listOf("Kotlin", "Java", "Python")  
Output:  
Kotlin, Java, Python

## Intermediate (40 Questions)

### **21. Find max element in a list**

Input:  
val nums = listOf(10, 50, 30, 70, 20)  
Output:  
70

### **22. Find min element in a list**

Input:  
val nums = listOf(10, 50, 30, 70, 20)  
Output:  
10

### **23. Count even numbers**

Input:  
val nums = listOf(1, 2, 3, 4, 5, 6)  
Output:  
3

### **24. Count odd numbers**

Input:  
val nums = listOf(1, 2, 3, 4, 5, 6)  
Output:  
3

### **25. Square each number**

Input:  
val nums = listOf(1, 2, 3, 4)  
Output:  
[1, 4, 9, 16]

### **26. Filter words starting with 'a'**

Input:  
val words = listOf("apple", "banana", "apricot", "cherry")  
Output:  
[apple, apricot]

### **27. Group words by first letter**

Input:  
val words = listOf("apple", "apricot", "banana", "blueberry")  
Output:  
{a=[apple, apricot], b=[banana, blueberry]}

### **28. Map words to lengths**

Input:  
val words = listOf("Kotlin", "Java", "Python")  
Output:  
[6, 4, 6]

### **29. Flatten list of lists**

Input:  
val data = listOf(listOf(1,2), listOf(3,4), listOf(5))  
Output:  
[1, 2, 3, 4, 5]

### **30. Distinct elements from list**

Input:  
val nums = listOf(1,2,2,3,4,4,5)  
Output:  
[1, 2, 3, 4, 5]

### **31. Sort strings by length**

Input:  
val words = listOf("sun", "moon", "star", "galaxy")  
Output:  
[sun, moon, star, galaxy]

### **32. Take first 3 elements**

Input:  
val nums = listOf(1,2,3,4,5)  
Output:  
[1,2,3]

### **33. Drop first 2 elements**

Input:  
val nums = listOf(1,2,3,4,5)  
Output:  
[3,4,5]

### **34. Check if all numbers > 0**

Input:  
val nums = listOf(1,2,3,-1)  
Output:

false

### **35. Check if any number is even**

Input:

```
val nums = listOf(1,3,5,8)
```

Output:

true

### **36. Zip two lists**

Input:

```
val a = listOf(1,2,3), b = listOf("a","b","c")
```

Output:

```
[(1,a), (2,b), (3,c)]
```

### **37. Unzip a list of pairs**

Input:

```
val data = listOf(1 to "a", 2 to "b", 3 to "c")
```

Output:

```
[(1,2,3), [a,b,c]]
```

### **38. Partition numbers into even/odd**

Input:

```
val nums = listOf(1,2,3,4,5,6)
```

Output:

```
[(2,4,6), [1,3,5]]
```

### **39. Fold sum of numbers**

Input:

```
val nums = listOf(1,2,3,4,5)
```

Output:

15

### **40. Reduce product of numbers**

Input:

```
val nums = listOf(2,3,4)
```

Output:

24

### **41. Find longest word**

Input:

```
val words = listOf("sky", "mountain", "forest", "butterfly")
```

Output:

butterfly

#### ***42. Find shortest word***

Input:

```
val words = listOf("sky", "mountain", "forest")
```

Output:

sky

#### ***43. Chunk list into size 2***

Input:

```
val nums = listOf(1,2,3,4,5)
```

Output:

```
[[1,2],[3,4],[5]]
```

#### ***44. Windowed sublists***

Input:

```
val nums = listOf(1,2,3,4)
```

Output:

```
[[1,2],[2,3],[3,4]]
```

#### ***45. Associate words with lengths***

Input:

```
val words = listOf("cat", "dog", "horse")
```

Output:

```
{cat=3, dog=3, horse=5}
```

#### ***46. Filter map values > 50***

Input:

```
val marks = mapOf("Alice" to 90, "Bob" to 40)
```

Output:

```
{Alice=90}
```

#### ***47. Get average of numbers***

Input:

```
val nums = listOf(10,20,30,40)
```

Output:

25.0

#### ***48. Find sum of even numbers***

Input:

```
val nums = listOf(1,2,3,4,5,6)
```

Output:

12

#### ***49. Find common elements of two sets***

Input:  
a = setOf(1,2,3), b = setOf(2,3,4)  
Output:  
[2,3]

### **50. Find union of two sets**

Input:  
a = setOf(1,2), b = setOf(2,3)  
Output:  
[1,2,3]

### **51. Find difference of two sets**

Input:  
a = setOf(1,2,3), b = setOf(2,3,4)  
Output:  
[1]

### **52. Shuffle list elements**

Input:  
val nums = listOf(1,2,3,4,5)  
Output:  
(random order like [3,1,5,4,2])

### **53. Find index of element**

Input:  
val nums = listOf(10,20,30,40)  
Find index of 30  
Output:  
2

### **54. Replace elements with \* if < 0**

Input:  
val nums = listOf(1,-2,3,-4)  
Output:  
[1,\*,3,\*]

### **55. Create frequency map of characters**

Input:  
val word = "banana"  
Output:  
{b=1, a=3, n=2}

### **56. Remove duplicates keeping order**

Input:  
val nums = listOf(1,2,2,3,1,4)



Output:  
[1,2,3,4]

### ***57. Pair elements with index***

Input:  
val nums = listOf(10,20,30)  
Output:  
[(0,10),(1,20),(2,30)]

### ***58. Find second largest number***

Input:  
val nums = listOf(10,20,50,40)  
Output:  
40

### ***59. Check palindrome list***

Input:  
val nums = listOf(1,2,3,2,1)  
Output:  
true

### ***60. Rotate list right by 2***

Input:  
val nums = listOf(1,2,3,4,5)  
Output:  
[4,5,1,2,3]